

SAN MIGUEL DE TUCUMAN, Argentina

WMO: 871210

Lat: 26.8372S

Lon: 65.1083W

Elev: 450

StdP: 96.03

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.3	4.9	-3.1	3.1	13.5	-1.4	3.6	11.8	8.0	16.3	6.9	14.8	2.9	320	0.368

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.2	36.7	23.5	35.1	23.4	33.8	23.2	26.5	32.7	25.7	31.8	25.0	31.0	4.6	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.8	21.0	30.8	24.0	19.9	29.8	23.2	18.9	28.7	86.2	33.0	82.6	31.8	79.2	31.1	31.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.4	7.0	5.9	0.6	40.4	1.7	1.5	-0.6	41.4	-1.5	42.3	-2.5	43.1	-3.7	44.2
			-0.4	28.2	1.8	1.0	-1.7	29.0	-2.7	29.5	-3.7	30.1	-5.0	30.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	20.4	25.8	24.8	23.0	20.1	16.7	13.8	13.4	16.1	19.0	22.2	24.2	25.5
	DBStd	5.57	2.89	3.11	3.05	3.37	3.33	2.89	3.55	4.20	4.41	4.04	3.61	3.19
	HDD10.0	18	0	0	0	0	1	4	9	3	1	0	0	0
	HDD18.3	542	0	1	4	20	72	137	157	95	43	12	2	1
	CDD10.0	3804	491	414	404	304	209	119	116	192	272	377	424	480
	CDD18.3	1286	232	181	149	74	22	2	4	26	64	131	177	222
	CDH23.3	11904	2248	1551	1041	429	108	11	54	337	749	1363	1792	2221
	CDH26.7	5150	1021	639	358	107	11	1	10	121	316	646	869	1052
Wind	WSAvg	2.9	3.1	3.0	2.7	2.5	2.4	2.5	2.8	3.0	3.2	3.2	3.2	3.2
Precipitation	PrecAvg	1018	223	184	167	66	27	13	9	9	16	65	104	164
	PrecMax	1504	451	388	392	152	98	37	30	39	73	162	196	375
	PrecMin	601	97	40	41	6	1	0	0	0	0	9	25	16
	PrecStd	220	98	78	84	39	23	8	8	12	15	41	51	81
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.5	36.2	34.0	31.2	28.1	24.9	27.8	34.0	35.9	38.2	38.2	38.8
		MCWB	25.2	25.7	25.1	23.6	20.9	18.3	18.3	20.1	21.1	22.7	23.1	24.1
	2%	DB	35.1	34.0	32.0	29.5	26.1	22.6	24.6	30.0	32.9	35.2	35.9	35.9
		MCWB	24.9	25.2	24.5	22.9	19.8	16.4	16.3	18.1	19.4	21.8	22.3	24.0
	5%	DB	33.5	32.2	30.2	27.8	24.2	21.0	22.5	27.0	30.4	32.9	33.8	34.0
		MCWB	24.4	24.7	23.6	21.7	18.7	15.0	14.9	16.7	18.2	20.6	22.0	23.9
	10%	DB	31.9	30.6	28.8	25.9	22.3	19.2	20.2	24.2	27.9	30.3	31.8	32.1
		MCWB	24.0	24.0	22.9	20.8	17.7	14.4	13.7	15.2	16.9	19.6	21.4	23.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.6	27.3	26.5	25.0	21.9	19.3	18.9	20.7	22.6	24.7	25.9	27.0
		MCDB	33.9	33.1	31.8	30.0	26.2	23.2	26.4	31.5	31.7	34.3	33.0	34.1
	2%	WB	26.3	26.3	25.3	23.6	20.7	17.6	17.3	19.0	20.8	23.1	24.6	25.8
		MCDB	32.6	32.0	30.4	27.8	24.5	21.1	23.1	28.5	30.3	32.0	32.0	32.8
	5%	WB	25.4	25.4	24.4	22.5	19.7	16.5	16.0	17.5	19.4	22.0	23.5	24.9
		MCDB	31.5	30.8	29.0	26.3	23.0	19.5	20.7	25.6	27.7	29.6	30.8	32.0
	10%	WB	24.6	24.5	23.4	21.5	18.6	15.3	14.5	16.0	18.1	20.8	22.5	24.0
		MCDB	30.3	29.6	27.5	24.9	21.7	18.2	19.0	23.0	26.2	27.9	29.3	30.4
Mean Daily Temperature Range	5% DB	MDBR	10.2	9.2	8.6	8.7	9.7	11.8	13.5	13.7	12.3	11.7	10.9	
		MCDBR	12.7	11.9	11.3	11.5	11.6	12.6	15.0	17.3	17.7	16.8	15.5	14.2
		MCWBR	5.7	5.6	5.7	6.1	6.4	7.2	8.0	8.3	7.7	6.7	6.4	6.2
	5% WB	MCDBR	11.4	10.8	10.2	9.8	9.8	10.3	13.0	15.8	15.4	14.0	12.7	12.3
		MCWBR	5.8	5.6	5.6	5.6	5.8	6.4	7.4	7.9	7.1	6.2	6.0	6.0
Clear-Sky Solar Irradiance	taub	0.408	0.407	0.392	0.370	0.331	0.318	0.318	0.361	0.429	0.433	0.414	0.413	
	taud	2.390	2.388	2.430	2.447	2.508	2.510	2.478	2.346	2.156	2.227	2.325	2.368	
	Ebn at Noon	936	920	904	876	871	862	876	871	853	885	924	934	
	Edh at Noon	129	126	115	103	88	83	89	111	148	146	136	132	
All-Sky Solar Radiation	RadAvg	6.29	5.75	4.83	4.03	3.35	3.15	3.65	4.57	5.48	5.93	6.35	6.27	
	RadStd	0.47	0.46	0.40	0.35	0.34	0.28	0.24	0.28	0.44	0.49	0.56	0.41	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	+0.95	+0.37	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	+0.76	+0.45	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page